
DeckForge AI

A locally fine-tuned diffusion model for skateboard decals,
on 8 GB of VRAM

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The assignment, and the problem underneath it

- A skateboard manufacturer wants **customer-designed decks**, without a designer in the loop for every order
- The assignment: a **self-trained** generator, **text + reference image**, an **interactive 3D deck**, deployed or demonstrably reproducible
- The hard part is not the pipeline. It is that **all of it has to run on one consumer laptop GPU**
- Assessed deliverable: the **research process**, not the artefact

docs/00-project-brief.md · thirteen mandatory requirements · learning outcomes D1–D7

The research question

How can a **locally fine-tuned** diffusion model, conditioned on **both a text prompt and a reference image**, generate skateboard-decal artwork in **multiple visually distinct styles** with **reproducible quality** on consumer hardware with **8 GB of VRAM**?

Twelve secondary research questions, RQ1–RQ12 · report §5

One constraint governs everything

8187.5 MiB of VRAM. One RTX 4060 Laptop GPU.

The shipped stack peaks at **5143.73 MiB** and leaves **200 MiB spare** under real serving — **2.4 % of the card.**

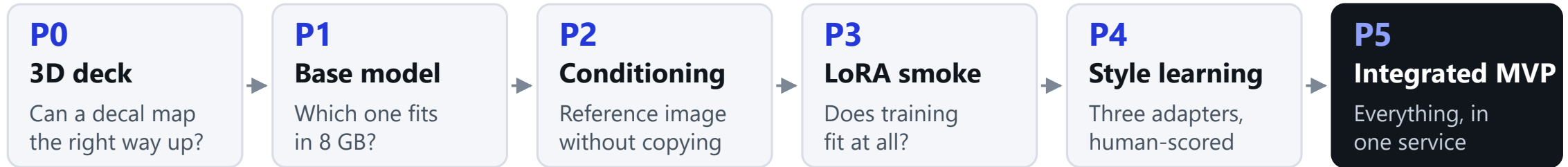
EXP-019b, EXP-032, EXP-034 · experiments/registry.csv · risk R12

How the work was done

- Every significant choice ran a **13-step loop**: question → alternatives → criteria → experiment → **measured** result → decision → validation
- **40 registered experiments**, each with its configuration, hardware readings and conclusion
- **16 decision records** — including the ones that rejected the option I expected to pick
- **Two human review gates** with a pre-declared 1–5 rubric, fixed seeds and blinding at the first gate
- Failed experiments stay in the write-up. **A deleted failure is a deleted result.**

[experiments/registry.csv](#) · [docs/decisions/](#) · [docs/05-experiment-methodology.md](#)

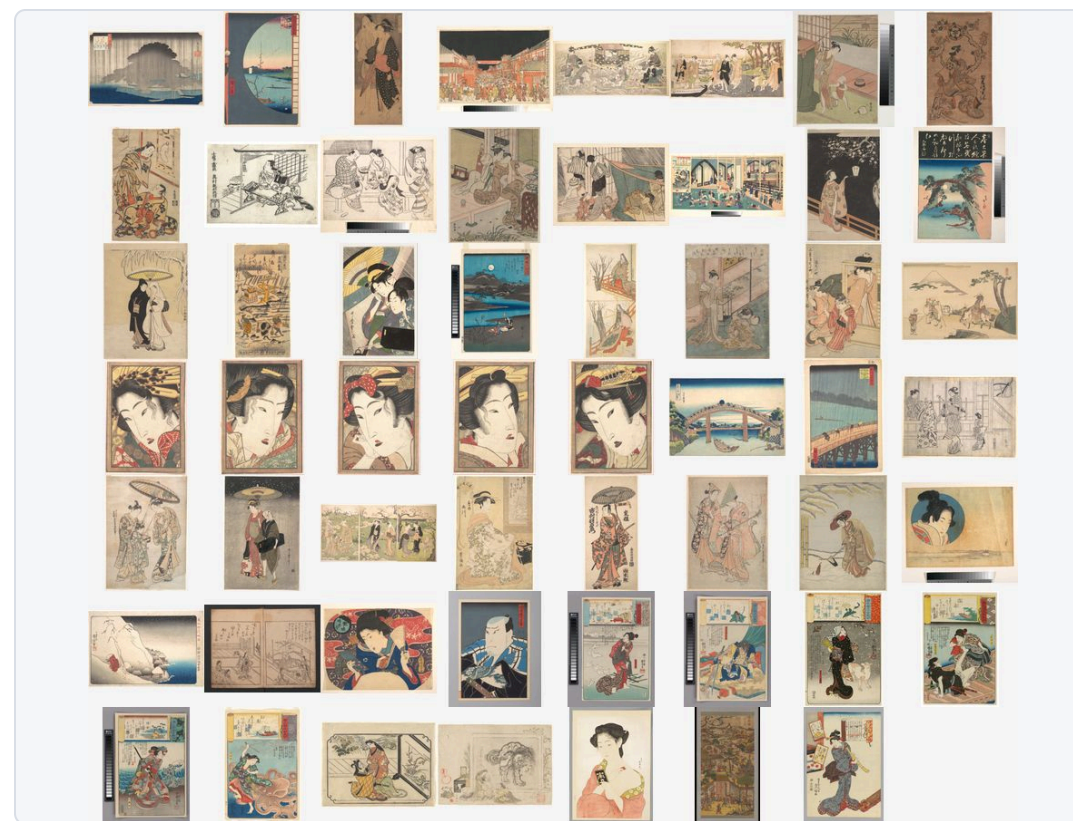
The prototype ladder



Each prototype answers the question the next one depends on. None was skipped, and P0 caught a geometry fault four milestones of convenient test

The dataset was built, not downloaded

- **148 items**, three styles, built for this project
- Ukiyo-e **55** · minimal-geometric **52** · retro-poster **41**
- Split **124 / 17 / 7** — the holdout was never seen in training
- Captions are **style-only**, not verbatim descriptions. That comparison was run, and style-only won



Ukiyo-e split. Metropolitan Museum of Art, CC0. Every item carries its licence and source URL in the manifest.

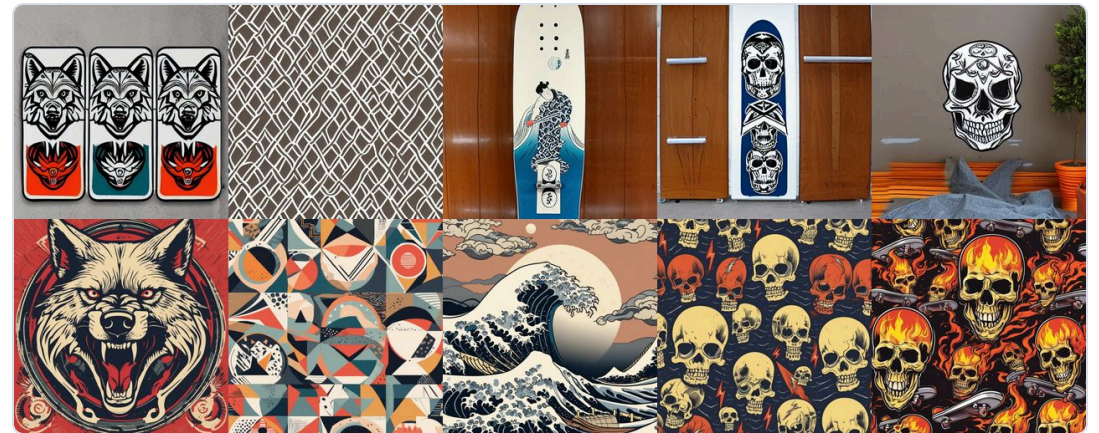
Licensing, decided before training

- Every item is **CC0, public domain, or created for this project**. There is no third-party licensed work in the training set
- **Licence and source URL recorded per item, in the manifest, before the item entered a split** — not audited afterwards
- Raw images are **git-ignored**; the manifest is committed. The evidence is the record, not the pixels
- Memorisation was **tested, not assumed** — nearest-neighbour checks against the holdout
- **What I cannot claim:** those checks bound copying, they do not prove it never happens

data/manifests/dataset-v1.csv · DR-006 · EXP-035 · report §10, §17

Base model: SDXL was better, and was rejected

- SDXL scored **better** than SD 1.5 on my own rubric, at its native resolution
- It was **rejected anyway**: 10 738 MiB allocated at 1024×1024, on a card with 8187.5 MiB
- Windows **spilled into host memory instead of failing**. All thirty runs reported success
- SD 1.5 selected — not because it wins on quality, but because **it is the one that fits at a useful resolution**



Track B — 1024×1024, SDXL's designed resolution. The images the 10 738 MiB measurement came from.

The habit that came out of it

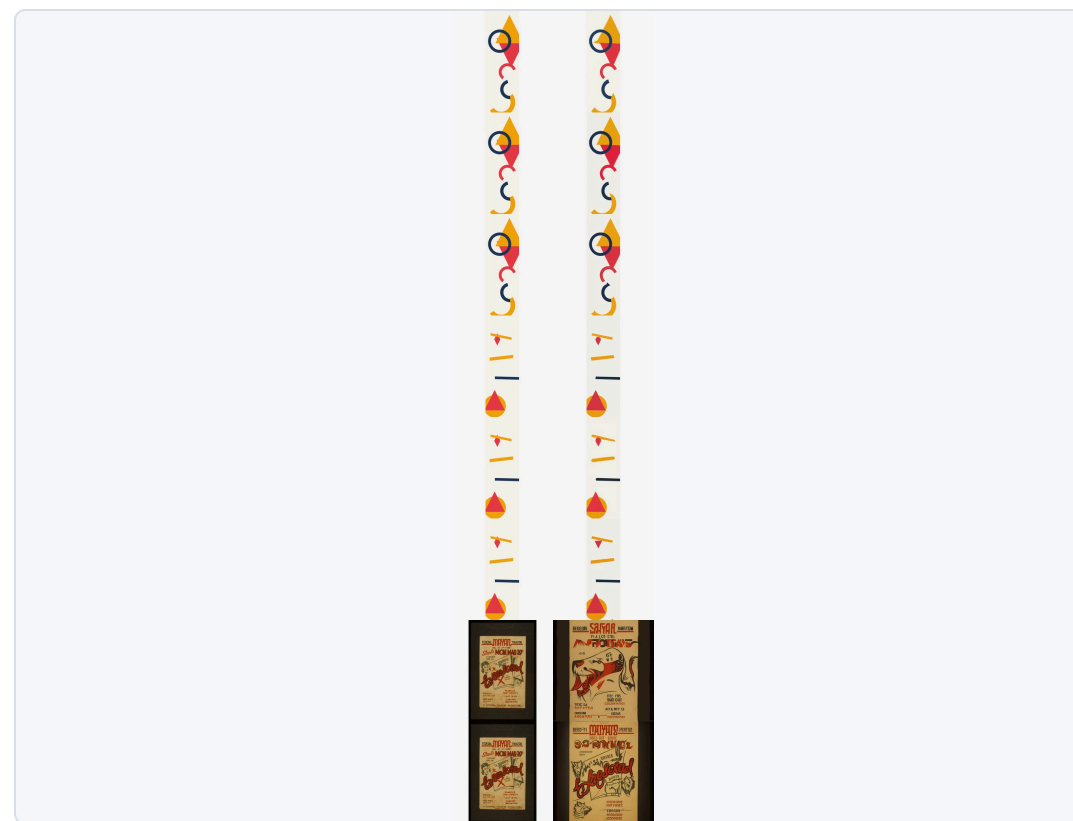
A successful run is not a run that fitted.

Read every memory figure **against the ceiling**, never against whether the process survived.

EXP-005 · the measurement-methodology correction · report §19.3

Reference conditioning: IP-Adapter over img2img

- Two ways to use a reference image: **img2img** or **IP-Adapter**
- img2img costs **zero extra VRAM** — decisive on this budget, and it lost anyway
- At the deck format it **returned its own input**. Every near-copy flag in that milestone came from img2img
- **IP-Adapter selected**, scale 0.55. A method that reproduces its input is not a generator.



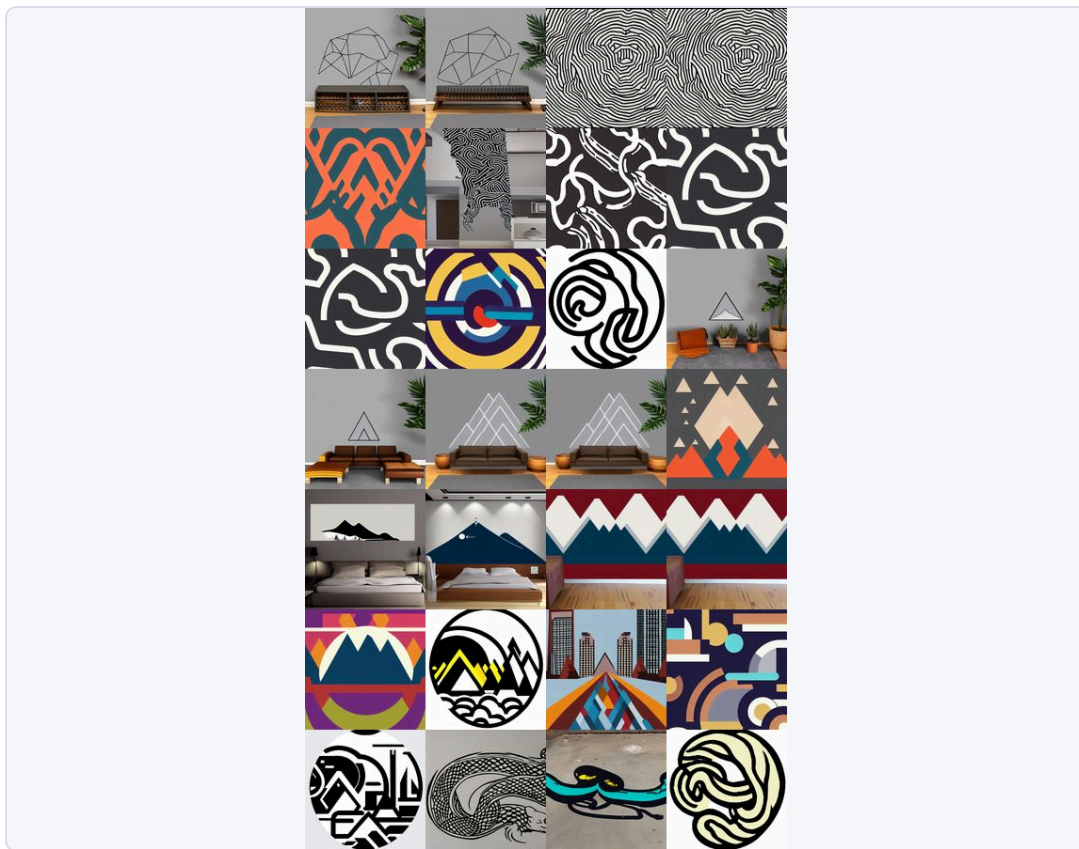
Reference, then img2img output. Some pairs are perceptually indistinguishable. This is why the free method was not selected.

Why LoRA, and what that does not claim

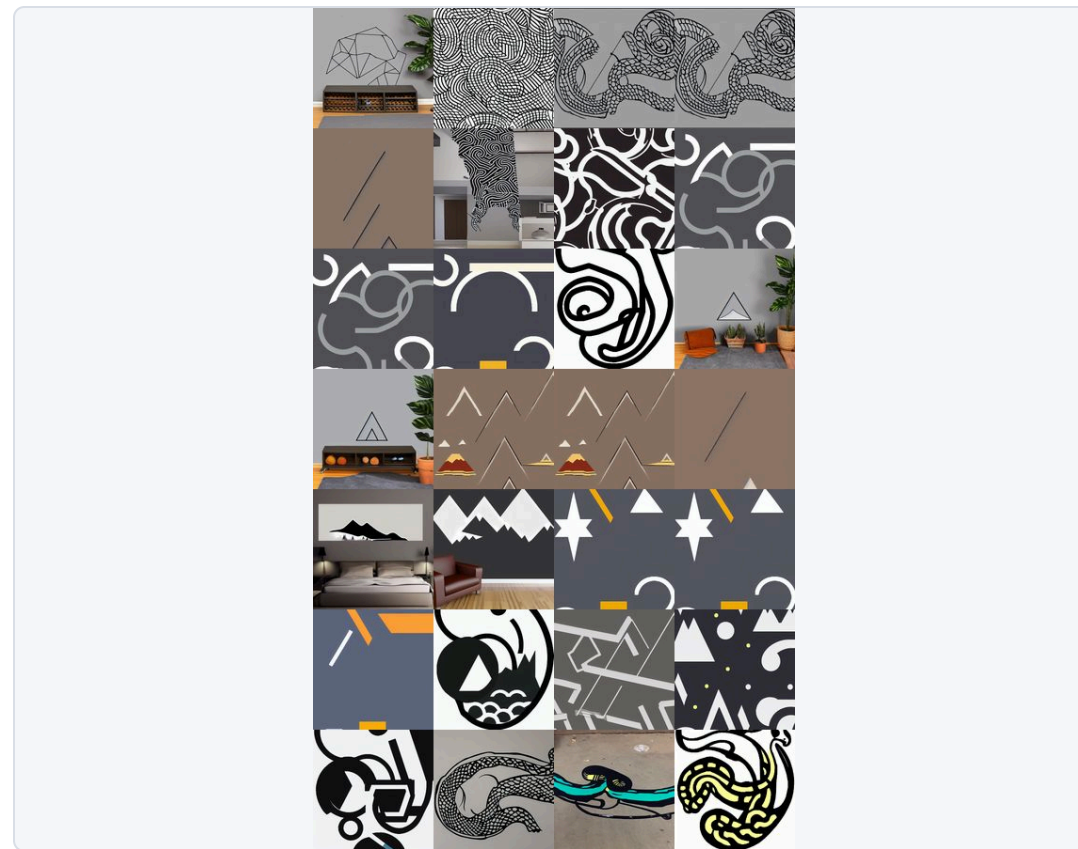
- Full fine-tuning does not fit. Training from scratch needs **millions of images and a cluster** — neither exists here
- LoRA trains a **small pair of low-rank matrices** beside the frozen base model
- Measured: **rank 8**, 3 133 MiB at 512^2 , 300 steps in **91 seconds**. It fits with room to work
- Each adapter is **6414480 bytes** — the styles ship as megabytes, not gigabytes
- **The claim is feasibility, not superiority.** Four alternatives were screened on criteria and never run

DR-009 · EXP-016 to EXP-019b · RQ1, recorded as bounded

Training longer made it less obedient



Step 300 — shipped. Style consistency 5, prompt adherence 4. EXP-027, square rubric set.



Step 600 — rejected. Style consistency still 5, prompt adherence 3. Same prompts, same seeds.

Training longer made the model more stylish and less obedient. Two of three styles ship at step 300. Ukiyo-e is the counterexample — it kept improving and ships at step 600.

What actually ships

STYLE	EXPERIMENT	SHIPS AT	VERDICT
minimal-geometric	EXP-027	step 300	PASS
ukiyo-e	EXP-028	step 600	PASS
retro-poster	EXP-029	step 300	PARTIAL PASS

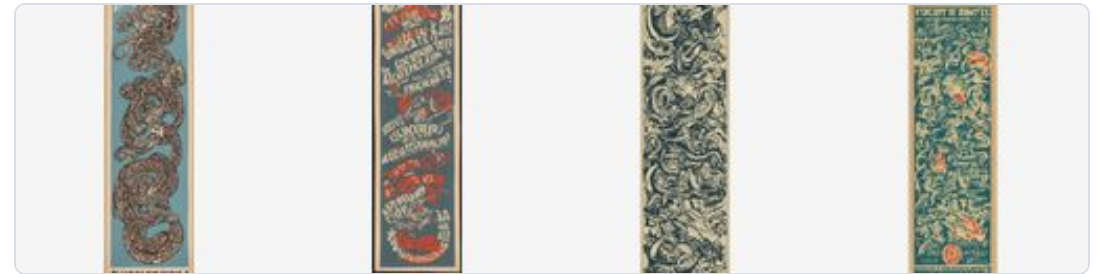
Applied at weight **0.7**. Every checkpoint is **verified by SHA-256 on every request** — the weights cannot be rebuilt, so they are treated as artifacts, not recipes.

docs/deployment/weights-manifest.md · guarded by pytest so the manifest cannot drift from the code

One style is a partial pass

retro-poster ships as a PARTIAL PASS.

- It learned the **frames, borders and display typography** of the archive posters, not only their style
- Shipped **with the limitation stated in the interface**, on every request
- Not dropped, not quietly upgraded, not re-scored until it passed



EXP-029, step 300. The frames and lettering are inherited from the source material, not requested by the prompt.

How many images does a style need?

How many images does a style need? I do not know. **INCONCLUSIVE**

- Trained the same style on **12, 24 and 44 images**, everything else held fixed
- The ordering came out **non-monotonic** — more images was not reliably better
- **No minimum count was established**, and none is claimed
- Reported as inconclusive rather than rounded into a recommendation

EXP-024n12, EXP-024n24, EXP-024 · RQ4, image-count component · report §13

The rest of what did not work

- **SD 2.1 could never be measured** — authentication-gated. The model decision rests on two candidates, not three
- **Memorisation is bounded, not settled.** The holdout checks limit copying; they do not disprove it
- **Style conditioning can dominate a detailed prompt** — ask for a skyline and a skateboarder and you may get neither, beautifully
- **Evaluation:** one human approver, **AI-assisted visual analysis**, and **no second independent human rater** — so no inter-rater agreement can be reported
- Institutional sources returned **HTTP 429 and 403** across five official paths; the wording that depended on them was removed

report §12 failed and blocked experiments · §18.4 validity threats

The integrated system

```
<rect x="404" y="40" width="360" height="200" rx="14" fill="#f4f6f9" stroke="#d9dee6" stroke-width="2"/>
```

```
<text x="428" y="80" font-size="24" font-weight="700" fill="#11151c">FastAPI service</text>
```

```
<text x="428" y="116" font-size="21" fill="#46525f">uvicorn, <tspan font-weight="700" fill="#11151c">FastAPI</tspan></text>
```

```
<text x="428" y="148" font-size="21" fill="#46525f">Busy lock &#8594; 409, never OOM</text>
```

```
<text x="428" y="180" font-size="21" fill="#46525f">Upload validation, timeouts</text>
```

```
<text x="428" y="206" font-size="21" fill="#46525f">SHA-256 checkpoint verify</text>
```

```
<rect x="834" y="40" width="380" height="200" rx="14" fill="#11151c"/>
```

```
<text x="858" y="80" font-size="24" font-weight="700" fill="#ffffff">One resident pipeline</text>
```

```
<text x="858" y="116" font-size="21" fill="#c8d0dc">Stable Diffusion 1.5</text>
```

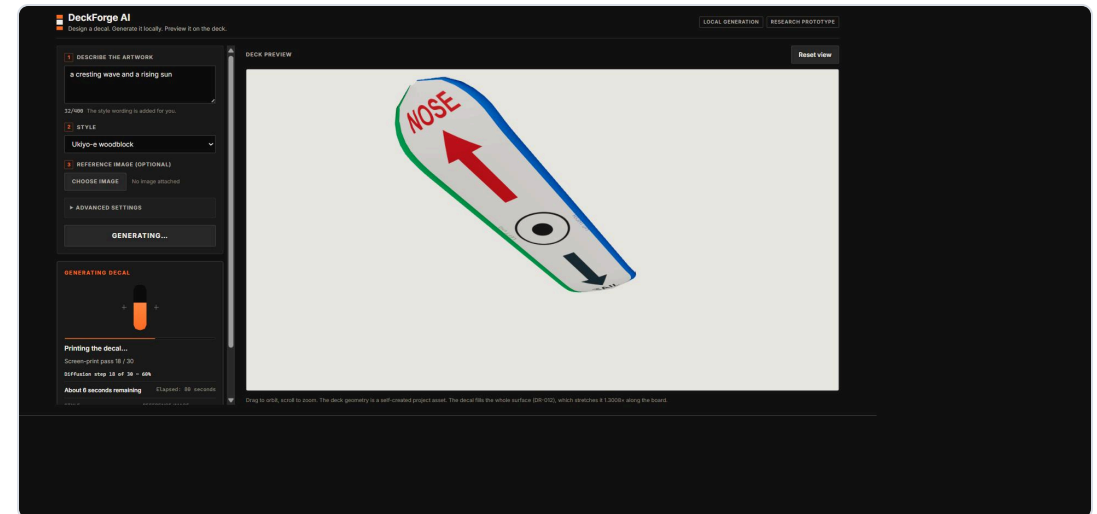
```
<text x="858" y="148" font-size="21" fill="#c8d0dc">+ one per-style LoRA</text>
```

```
<text x="858" y="180" font-size="21" fill="#c8d0dc">+ IP-Adapter (optional)</text>
```

```
<text x="858" y="206" font-size="21" fill="#8fa0ff">512 &#215; 1536, direct</text>
```

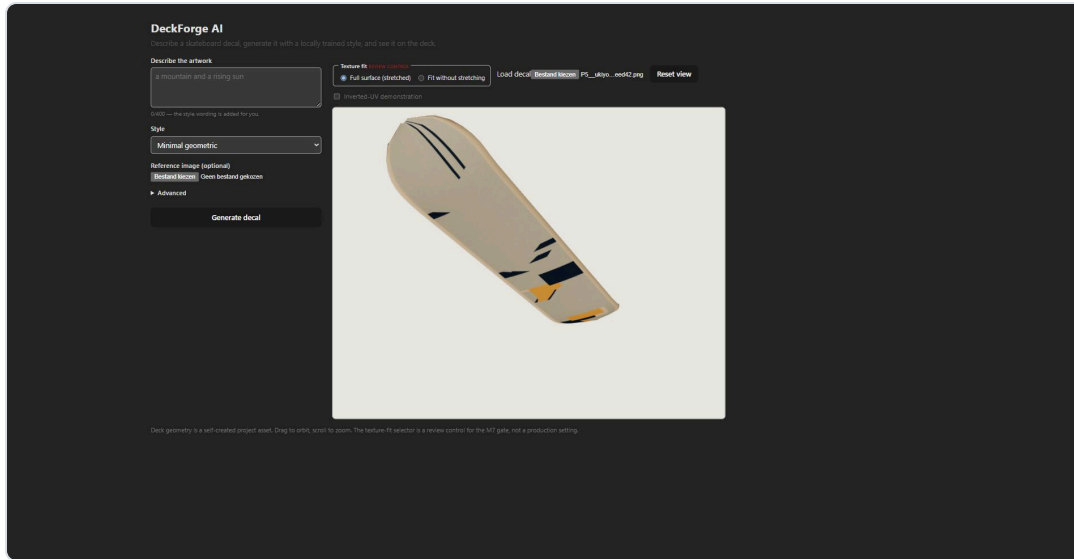
The MVP, on screen

- Generation is **direct to the deck format**, 512×1536 — not square-then-crop
- The progress bar reports **real diffusion steps**, from the pipeline
- While the model loads it shows **no percentage at all**, because loading exposes none. A bar creeping to 90 % would look better and be a lie
- Every result carries its **seed, pinned model revision, adapter hash and measured VRAM**

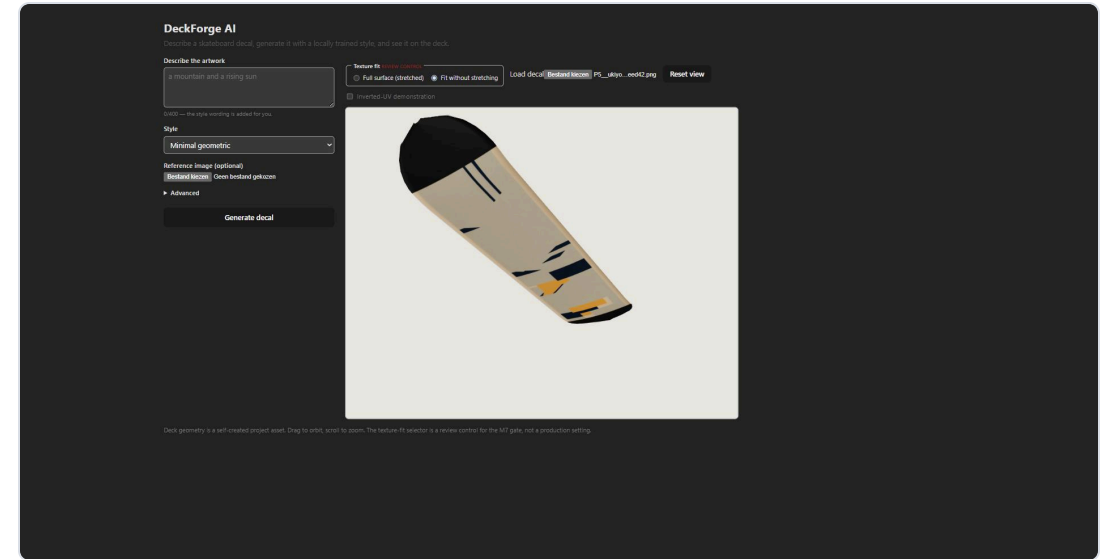


Denoising, step-accurate. The hypothesis that generating the deck ratio directly would degrade quality was tested, and refuted.

A trade-off the interface admits to



Fill the surface — shipped. The artwork is stretched 1.3×.



Preserve the ratio — rejected. 23 % of the board stays bare.

Decals are 1:3. The deck surface is 1:3.9. **Both options were built and compared.** The interface tells the user which one it chose, and that the artwork is stretched.

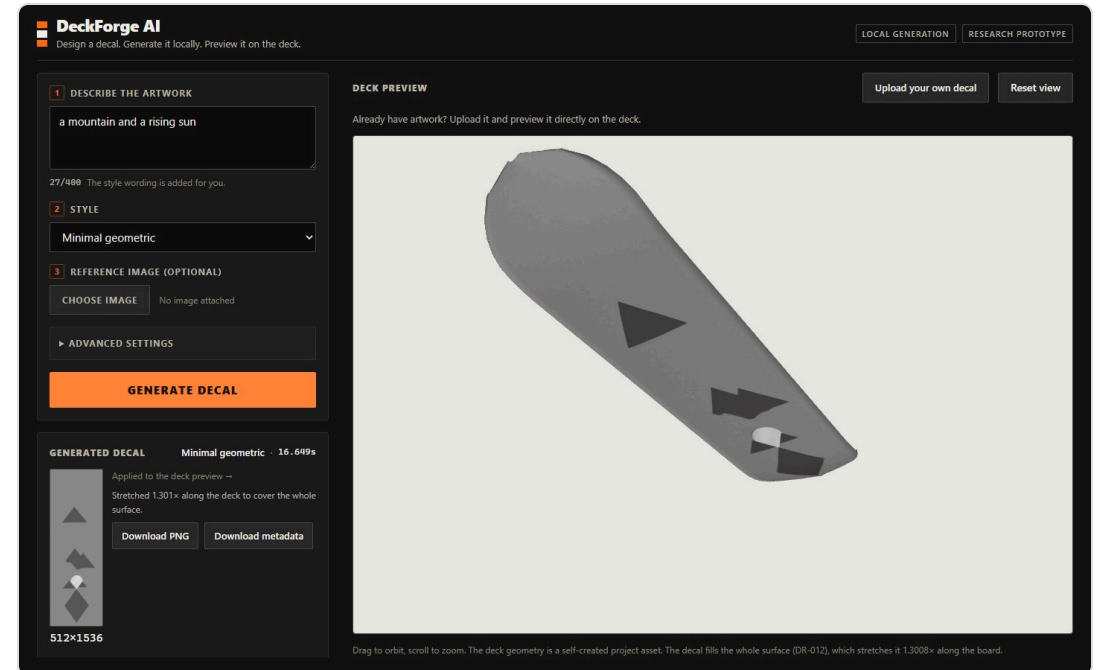
Testing

- **473** product and research tests + **16** report-validation tests = **489 pytest**
- **183** frontend unit tests · **38** browser end-to-end scenarios
- The **clean-clone test** is the one that counts: clone into an empty directory, install from scratch, reach a working system
- It caught an **integrity control that had only ever passed on the machine that wrote it**
- A green suite does not prove the generator makes good pictures. **Only the human gates did that**

[docs/evidence/M8/](#) · [docs/evidence/M8/clean-clone/](#) · [report §15](#)

Determinism has halves

- **Inference is deterministic.** A clean clone reproduced an earlier output **byte for byte**, three days later, in a freshly built environment
- **Training is not.** The adapter initialisation draws from an **unseeded global generator**, so a run cannot be repeated from its recorded seed
- So the checkpoints are **artifacts, verified by hash** — not recipes
- A defect that produces perfectly valid-looking runs, and is invisible unless you compare weights



Clean clone, fresh environment. SHA-256

46bbf160e4270429e6692467dc6c59577e99bf3178dedd8d38193d0335fb6d7f,

identical to the original run.

Deployment: local, and chosen

- Four options were compared. **Selected: native local run + pre-generated backup demo assets** (Option D of four)
- **200 MiB spare prevents a second resident pipeline on the validated GPU**, so multi-worker inference is unsupported
- Docker with GPU passthrough was **never verified on this machine**; its VRAM overhead is unmeasured
- **Public cloud deployment was deliberately not selected for this bachelor version** — it is a cost and a network dependency on demo day, for no research gain
- Reproducibility is carried by the **runbook and the clean-clone proof**, not by a public URL

DR-014 · alternatives A–D compared on nine criteria · EXP-034

Conclusions

Yes — and the configuration that fits is now specific and measured. SD 1.5 at 512×1536, one per-style LoRA at 0.7, IP-Adapter at 0.55, one resident pipeline.

Bounded three ways: **the margin is 2.4 % · reproducible describes inference, not training · "multiple distinct styles" is three, and one is partial.**

Not concluded: that LoRA beats the alternatives · that SD 1.5 is the best base model · that three styles is enough · that any image count is a threshold · that a green suite means good pictures.

report §19 · eight RQs answered in scope, four bounded, one component inconclusive

Live demo

Prompt → style → **512×1536 decal** → 3D deck. Running on this laptop's GPU. Nothing leaves the machine.

If the GPU does not cooperate, that is a live-demo problem and not a project problem — there is a recorded run ready and I will say which one you are looking at.

Learning outcomes, and what I would change

- **D1 / D5 / D6** — 40 experiments, six prototypes, conclusions bounded by what was measured
- **D3 / D2** — 16 decision records, a planning log that records what changed and why, public issues and board
- **D4** — alternatives compared and rejected on criteria: base model, conditioning, fine-tuning, deployment
- **D7** — a 91-page report, this deck, and a system that states its own limitations to its user
- **What I would change:** repeat every condition before drawing a curve · seed everything and verify by comparing weights, not by trusting the log · get a second rater

github.com/KylianAlgoet/Selftrained-and-deployed-AI-image-generator · [docs/learning-outcome-traceability.md](#)